MIL-G-52889C
1 July 1981
SUPERSEDING
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MILITARY SPECIFICATION

GENERATOR SETS, DIESEL ENGINE DRIVEN

5 KW, 60 HERTZ AND 10 KW, 60 AND 400 HERTZ TACTICAL, UTILITY,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the military-design, tactical-utility, air-cooled, diesel-engine (DED) generator sets (hereinafter referred to as "sets" or "set"), 5 kilowatt (kW), 60 Hertz (Hz) and 10 kW, 60 Hz and 400 Hz. The general characteristics of the sets are described in MIL-STD-633.

1.2 Classification. The sets shall be type I (tactical) class 2C (utility), mode II (400 Hz), mode III (60 Hz), and of the following sizes, in accordance with MIL-STD-1332, as specified (see 6.2):

Size 5 - 5 kW (MIL-G-52889/1) (60 Hz) MEP-002A
Size 10 - 10 kW (MIL-G-52889/2) (60 Hz) MEP-003A
Size 10 - 10 kW (MIL-G-52889/3) (400 Hz) MEP-112A

1.3 Voltage connections and rating. The sets shall be rated 0.8 power factor (pf) lagging and shall be reconnectable for the following voltage ratings:

(a) 1 phase, 2 wire, 120 volts.
(b) 1 phase, 3 wire, 120/240 volts.
(c) 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
W-R-550	- Rod, Ground (with attachments).

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-G-28554	- Generator Sets, Mobile Electric Power; Packaging of.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

SPECIFICATION SHEETS

MIL-G-52889/1	- Generator Sets, Diesel Engine Driven 5 kW, 60 Hz; Tactical, Utility.
MIL-G-52889/2	- Generator Sets, Diesel Engine Driven 10 kW, 60 Hz; Tactical, Utility.
MIL-G-52889/3	- Generator Sets, Diesel Engine Driven 10 kW, 400 Hz; Tactical, Utility.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.

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MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-705	- Generator Sets, Engine Driven, Methods of Tests and Instructions.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MILITARY

MIL-HDBK-705	- Generator Sets, Electrical, Measurements and Instrumentation.

2.1.2 Other government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

TA 30554-72-5984	- Slave Receptacle Assembly

(Copies of specifications, standards, drawings, and handbook required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail requirements for individual set sizes. The individual set requirements shall be as specified herein and in the applicable specification sheets. In the event of any conflict among the requirements thereof, the order of precedence shall be the detail specification, the general specification, and the drawings.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The contractor shall furnish one generator with exciter-regulator system and two complete assembled sets for examination and tests within the time frame specified (see 6.2) to prove, prior to starting production, that he is capable of producing sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor, and shall be subject to surveillance and approval by the Government (see 6.5). When specified (see 6.2), two additional preproduction model sets shall be furnished for performing the reliability test (see 3.7 and 4.9.3).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more sets for inspection as specified in 4.6.

3.4 Starting, operating, handling, and storage.

3.4.1 Starting and operating. When equipped with all accessories, the set shall start (see 6.4.1) and operate within its rating as follows.

3.4.1.1 Starting.

3.4.1.1.1 Without winterization system. Within 5 minutes, using set batteries, the set shall start as follows:

(a) At ambient temperatures from plus 125° F (51.7° C) to minus 25° F (minus 31.7° C) at all possible relative humidity (rh).
(b) With 12 inches of rain per hour impinging on the set at angles from the vertical up to 45 degrees from all directions.
(c) With base of the set in planes from level to 15 degrees from level.
(d) At any altitude up to and including 8000 feet above sea level (564.9 millimeters of mercury (mm Hg)) at ambient temperature of 95° F (35° C).

3.4.1.1.2 With winterization system. When the winterization system is used at ambient temperature from minus 25° F (minus 31.7° C) to minus 50° F (minus 45.6° C), the set shall start as follows:

(a) Within 60 minutes, with use of the fuel-burning winterization system for a maximum of 55 minutes.
(b) Within 5 minutes, after a 24 hour period of standby operation (see 6.4.2) using the fuel-burning winterization system.

3.4.1.2 Operating. After starting, and with a warmup period of not more than 15 minutes at no load, the set shall operate in accordance with the provisions specified herein at continuous loads up to and including rated load (see 6.4.3) under the conditions specified in 3.4.1.1, except for the altitude condition (see 3.4.1.1.1(d)) the set shall meet the above conditions at rated load for 5000 feet (632.4 mm Hg) at 107° F (41.7° C), and 90 percent of rated load at

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8000 feet (564.9 mm Hg) at 95° F (35° C). Maintenance shall be limited to the scheduled (preventative) maintenance permitted by Table III and unscheduled (corrective) maintenance as permitted by the reliability requirements of 3.7 and the valve adjustment requirement of 3.8.

3.4.2 Operating speeds. Operating (rated) speed of the set shall be 1800 revolutions per minute (rpm) (60 Hz) and 2000 rpm (400 Hz). Sets shall be capable of operating at 2500 rpm for a period of 5 minutes without damage.

3.4.3 Rough handling. The set shall not be damaged (see 6.4.5) by rough handling which could be encountered during rail, truck, trailer, aircraft, or helicopter transportation (see 6.4.6, 6.4.7 and 6.4.8).

3.4.4 Storage temperature and humidity. The set shall not be damaged (see 6.4.4) by storage in temperatures from plus 155° F (plus 68.3° C) to minus 65° F (minus 53.9° C) at all possible relative humidity for temperature within that range and shall not be damaged by exposure to five consecutive 48 hour humidity cycles as specified in MIL-STD-705, method 711.1.

3.4.5 Lifting and towing provisions. The lifting attachment shall be capable of withstanding not less than eight times the total weight of the set. The set shall remain within 15 degrees of level when lifted by the lifting attachment. The towing provisions shall be capable of withstanding a pull of not less than five times the total weight of the set.

3.4.6 Flexural vibrations and critical speeds. The set shall be free from dangerous flexural vibration and dangerous torsional critical speeds (see 6.4.16 and 6.4.17) between 1400 and 2500 RPM.

3.5 Maximum power. The maximum output (corrected) in kW for each set shall be not less than 125 percent of rated load when operated in the 3 phase connection, rated frequency, rated voltage, and at unity power factor.

3.6 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the set shall not exceed the UM04 limits of class C2 equipment of MIL-STD-461.

3.7 Reliability/Maintainability. The set shall have an upper test mean time between failure (specified MTBF) of 500 hours and a lower test mean time between failure (minimum acceptable MTBF) of 250 hours. The maintenance ratio (see 6.4.18) shall not be more than 0.04.

3.8 Engine. The engine, including all systems, components, accessories, and auxiliaries, shall conform to the applicable drawings and shall meet the requirements as specified herein. After the initial 50 hours of operation each engine shall have its valve tappet clearances adjusted.

3.8.1 Oil temperature. The oil temperature measured at the bottom of the oil dipstick shall not exceed 275° F when operated under any of the conditions specified in 3.4.1.

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3.8.2 Fuels. The set, when operating on DF-1, DF-2 or DF-A diesel fuel conforming to VV-F-800, shall meet the requirements specified herein at all loads up to and including rated load. All components of the set and fuel system shall be capable of operating on JP-4 turbine fuel conforming to MIL-T-5624 continuously for at least 200 hours. When operating on JP-4 turbine fuel, the set shall meet all requirements herein except as follows:

(a) The set shall be capable of starting and operating at ambient temperatures from 0° F (minus 17.8° C) to plus 100° F (37.8° C).
(b) Rated load may be reduced not more than 15 percent.

3.8.3 Governing system performance. The governing system, when properly adjusted, shall perform as specified herein.

3.8.3.1 Frequency regulation. The govering system shall maintain the frequency (see 6.4.9) within 3 percent of rated frequency for all loads from no load up to and including rated load.

3.8.3.2 Short-term steady-state stability (30 seconds). At every constant load from no-load to rated load (see 6.4.3), the system shall maintain frequency within a bandwidth equal to 2 percent of rated frequency. The system shall not permit repetitive frequency variations, commonly called "hunting", even though within the allowable 2 percent band.

3.8.3.3 Long-term steady-state stability (4 hours). At constant ambient temperature, constant barometric pressure, constant voltage and any constant load from no-load to rated load, the system shall maintain frequency within a bandwidth equal to 3 percent of rated frequency in a 4 hour operation period.

3.8.3.4 Frequency drift. With the regulation set at not more than 3.0 percent and with the set operating at constant load and voltage, a change in ambient temperature of up to 60 F° (33.3 C°) in an 8-hour period shall not cause the frequency to change by more than 2 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.8.3.5 Transient performance. Following any sudden increase in load, including from no load to rated load, the governing system shall reestablish stable engine operating conditions (see 6.4.10) within 3 seconds, and the maximum transient frequency change below the new steady-state frequency (undershoot) shall be not more than 3 percent of rated frequency. Following any sudden decrease in load, including from rated load to no load, the governing system shall reestablish stable engine operating condition within 3 seconds, and the maximum transient frequency change above the new steady-state frequency (overshoot) shall be not more than 4 percent of rated frequency.

3.8.3.6 Frequency adjustment range. The manual speed control shall permit a frequency adjustment of plus or minus 3 percent of rated frequency at all load conditions from no load to rated load. A 90 degree turn of the knob shall not cause a change in frequency of more than 1 percent of rated frequency. Clockwise rotation of vernier adjustment shall cause the frequency to increase.

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3.8.4 Engine protective devices. The set shall be equipped with the following engine protective devices.

3.8.4.1 Engine over-temperature. The engine over-temperature protective device shall not impede set starting and operating under the conditions cited in 3.4.1 and shall shut down the engine before the surface temperature at the corner of the switch mounting boss adjacent to the switch reaches 470° F (243.3° C).

3.8.4.2 Low lubrication oil pressure. The engine low lubrication oil pressure device shall shut down the engine when the oil pressure is 12 to 15 pounds per square inch.

3.9 Generator.

3.9.1 Generator rotor assembly. The rotor assembly shall be in electrical and mechanical balance at all speeds up to 2500 rpm and shall withstand operation at 2500 rpm for not less than 15 minutes.

3.9.2 Winding resistance, stator and field winding. The winding resistance of each winding shall not differ by more than plus or minus 10 percent from the resistance specified on the applicable drawing. All resistance values shall be corrected to a temperature of 77° F (25° C).

3.9.3 Insulation resistance. The insulation resistance of all circuits shall be not less than 1 megohm when the circuits are stabilized at an ambient temperature of 77° F (25° C) plus or minus 9 F° (5 C°).

3.9.4 Temperature rise. The maximum temperature rise, with the set delivering rated load continuously, shall be not more than 180 F° (100 C°) for the stator windings and connections when measured by the rise in resistance method.

3.9.5 Dielectric strength. Electric windings shall withstand the following 60 Hz voltage applied for 1 minute:

(a) Generator armature: 1480 V between phase windings, and 1240 V between phase windings and ground.
(b) Generator field and exciter windings: Ten times ceiling voltage but not less than 1500 V nor more than 3500 V (applied between windings and ground).
(c) Windings energized by the 24 V dc control, cranking and battery charging system: 500 V (applied between the winding and ground.
(d) All others: Twice rated voltage plus 1000 V (applied between the windings and ground or between windings, as applicable).

3.9.6 Short circuit. The generator, exciter and voltage regulator operating as a unit shall withstand the conditions as specified in the detailed specification (see 3.1).

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3.9.7 Transient reactance, rated voltage. The direct axis transient reactance shall not exceed that specified in the detailed specification (see 3.1).

3.9.8 Negative sequence impedance. The generator negative sequence impedance shall not exceed that specified in the detailed specification (see 3.1).

3.9.9 Voltage waveform. The deviation factor and any single voltage harmonic for the line-to-line and line-to-neutral voltage for each voltage connection and load connection shall be not more than as follows:

(a) The deviation factor shall not exceed 6 percent in the single phase connections or 5 percent in the 3 phase, 120/208 volt connection.
(b) Single frequency harmonics shall not exceed 3 percent in the single phase connections or 2 percent in the 3 phase, 120/208 volt connection.

In addition, there shall be no evident discontinuities, spikes, or notches in the waveform when viewed on an oscilloscope having a bandwidth of DC to 1.5 mHz and useable viewing screen of 8 x 10 cm. The oscilloscope gain shall be adjusted such that one cycle of voltage wave covers approximately the entire viewing screen.

3.9.10 Voltage unbalance. Maximum difference between line-to-line voltages shall be not more than 5 percent of rated voltage for the 120/208 volt connection under the conditions of a single phase, line-to-line unity pf load of 25 percent of rated current and no other load on the generator.

3.9.11 Phase balance voltage. With the set operating at rated voltage and frequency and no load, the maximum difference in the three line-to-neutral voltages shall be not more than 1 percent of the rated line-to-neutral voltage when connected for 120/208 volts, 3 phase, 4 wire. The maximum difference between the voltage of the 120-volt windings of any phase shall not exceed 1 volt.

3.9.12 Phase sequence (rotation). Phase rotation shall be L_1, L_2 and L_3.

3.9.13 Voltage performance. The generator, together with the regulation systems, shall automatically keep the voltage within the limits specified herein, when measured with true root mean square (rms) sensing and indicating meters.

3.9.13.1 Voltage regulation. The voltage regulation from no load to rated load and from rated load to no load shall be not more than 3 percent of the rated voltage for all voltage connections throughout the voltage ranges specified in 3.9.13.7.

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3.9.13.2 Voltage short-term stability (30 seconds). At every constant load from no load to rated load, the voltage at the set terminals shall remain within a bandwidth equal to 2 percent of rated voltage.

3.9.13.3 Voltage long-term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant frequency and any constant load from no load to rated load, the voltage shall remain within a bandwidth of 4 percent of rated voltage in a 4-hour operating period.

3.9.13.4 Voltage drift. With the set operating at constant load and frequency, a change in ambient temperature up to 60 F° (33.3 C°) in an 8-hour period shall not cause the voltage to change by more than 2 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.9.13.5 Voltage dip and rise for rated load. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows: With the set initially operating at rated frequency and rated voltage and following any sudden change in load from no load to rated load, the instantaneous terminal voltage shall not dip more than 20 percent of rated voltage and shall reach stable conditions (see 6.4.10) within 3 seconds. Overshoot or undershoot of the final voltage shall not exceed the initial voltage transient in amplitudes. The above requirements shall also apply when the load is suddenly changed from rated load to no load, except that the initial voltage transient shall involve a voltage rise not to exceed 20 percent of rated voltage.

3.9.13.6 Voltage dip for low power factor load. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows: With the set operating at no load and the corresponding voltage and frequency conditions, the output voltage of the set shall not dip more than 35 percent of the no load voltage when a balanced 3 phase 0.4 pf (or less) lagging static load of twice rated set current (1/2 per unit impedance) is suddenly applied to the set output terminals. The output voltage shall recover to a minimum of 95 percent of the no load voltage within 5 seconds and shall stabilize at or above this voltage.

3.9.13.7 Voltage adjustment range. The minimum voltage adjustment range shall be:

(a) 205 to 220 volts for the 120/208 volt connection.
(b) 114 to 126 volts for the 120 volt connection.
(c) 228 to 252 volts for the 240 volt connection.

This voltage adjustment range shall apply for all loads between no load and rated kW, rated power factor at rated frequency under all conditions of 3.4.1. Clockwise rotation of the voltage adjustment control shall increase the set output voltage.

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3.9.13.8 Voltage modulation. The voltage modulation of the output voltage for any load up to and including rated load shall not exceed that specified in the detailed specification (see 3.1.) This requirement shall apply to all line-to-line and line-to-neutral voltages.

3.9.14 Field currents. The generator exciter field current of each individual generator for the 3 phase, 120/208 volt connection at rated frequency and normal ambient conditions (see 6.4.13) shall not exceed the values specified in the detailed specification (see 3.1).

3.9.15 Generator efficiency. The generator assembly, including the excitation system shall have a minimum efficiency at normal ambient temperature (see 6.4.13) as specified in the detailed specification (see 3.1).

3.10 Alternating current circuit interrupter. The alternating current circuit interrupter shall operate as indicated in the detailed specification (see 3.1) for all voltage connections.

3.11 Control system. All dc control devices shall be suitable for operation on voltages of 20 to 32 volts. Total dc current for all control devices shall not exceed 5 amperes when the set is operating. The control system circuit breaker and reverse battery polarity device shall prevent damage to the control circuits and alternator in the event of reversed battery polarity. When the set is not operating with the master switch in the "OFF" position, no current drain in excess of 5 milliamps shall be imposed upon the batteries except that for the panel lights, when the associated switch is in the "ON" position.

3.12 Instruments. The allowable error (see 6.4.11) of each metering system, including the electrical circuit, under all conditions specified in 3.4.1, shall be as specified herein.

3.12.1 Voltmeter system. The error of the voltmeter system shall be not more than 9 volts.

3.12.2 Ammeter system. The error of the ammeter system shall be not more than as follows:

True percentage load	Set meter indication (limits)
25	15-35
50	40-60
75	65-85
100	90-110

3.12.3 Frequency meter system. The error of the frequency meter system shall not be more than 1.5 Hz for 60 Hz operation and 6 Hz for 400 Hz operation.

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3.13 Delivery connection tag. Unless otherwise specified (see 6.2), each set shall be connected for 120/208 volts, 3 phase and shall have a connection tag attached to one of the load terminals indicating: "This set is connected for 120/208 volts. See connection changeover instructions for other voltage ratings".

3.14 Audio noise. Audio noise sound pressure levels emanating from the set when it is operating at any load up to and including rated load shall not exceed the values shown below, when the microphone is located 5 feet 8 inches above the ground on a 25-foot radius circle measured in any direction from the geometrical center of the set:

Octave band center Frequencies (Hz)	Maximum Acceptable Noise Level in Decibels (0.0002 Microbar Reference)
63	95
125	95
250	95
500	85
1000	80
2000	80
4000	75
8000	70

If pure tones, or narrow bands of noise, are present in any octave band, the sound pressure level permissible for that octave shall be reduced from the above values by 5 decibels (db) for frequencies above 1,000 Hz and 10 db for frequencies below 1,000 Hz.

3.15 Slave receptacle. When specified (see 6.2), a receptacle assembly in accordance with TA 30554-72-5984, shall be supplied, mounted and connected.

3.16 Ground rod assembly. When specified (see 6.2), a ground rod assembly in accordance with W-R-550, shall be supplied.

3.17 Treatment and painting. The set shall be treated and painted in accordance with the applicable drawings (see 3.1), except as specified herein. The following shall not be painted: terminal wiring connections, governor linkage, instruction diagrams, nameplates, rectifiers, relays, circuit breakers, switches, hose, and all other parts whose operation or function would be adversely affected by paint. Treatment and painting shall be optional for the following if located within the set housing and not visible during operation of the set: corrosion-resisting metals; metals treated to be corrosion resistant; and items treated and painted to be corrosion resistant by the original manufacturers. After assembly and testing, the interior of the set, the entire exterior of the set, except items adversely affected, shall be oversprayed.

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3.18 Workmanship, safety, and human engineering. Workmanship, safety, and human engineering shall be in accordance with MIL-STD-454, requirements 9, 1, and 62 respectively. In addition, workmanship shall be of a quality to assure delivery of sets which are free from defects resulting from defective material and improper manufacturing or assembly practices. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Components and material inspection. The contractor is responsible for insuring that components and material used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP insection, the sampling plan, and the basis for acceptance and rejection (see 6.4.14 and 6.6)

4.1.3 Disassembly inspection. Failure of any examination or test by the first article (preproduction model) shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article (preproduction model). Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.4 Inspection failure. Should the Government elect to perform or repeat any inspection or test in this specification, failure of a set to meet any requirement specified herein shall be cause for refusal by the Government to accept production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on any sets previously produced under the contract. Any deficiencies

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found as the result of such elective inspection or test will be considered prima facie evidence that all sets accepted prior to the completion of such inspection or test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. The provisions of this paragraph apply, notwithstanding any prior acceptance of first articles or first article test reports.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (preproduction) inspection (see 4.3).
(b) Individual generator inspection (see 4.4).
(c) Individual set inspection (see 4.5).
(d) Initial production inspection (see 4.6).
(e) Quality conformance inspection (see 4.7).
(f) Inspection comparison (see 4.10).
(g) Inspection of packaging (see 4.11).

4.3 First article (preproduction) inspection.

4.3.1 Examination of components and subassemblies. Examination of components and subassemblies shall be made prior to assembly of the generator or set. Evidence that any components or subassemblies do not comply with the requirements of the drawings shall be cause for rejection of that component or subassembly.

4.3.2 Examination of generator and sets. Examination of the generator (with exciter-regulator system) and the sets shall be made without disassembly. Evidence that the generator (with exciter-regulator system) or sets do not comply with the drawings shall be cause for rejection of the generator or sets.

4.3.3 Tests. The generator with exciter-regulator system and the sets shall be subjected to the tests marked "X" in column 1 of table I. The set tests marked "X" under column 1 of table I may be conducted in any order, except that following the "insulation resistance test" the first three set tests shall be "voltage modulation", "railroad impact", and "humidity" in that order, and the last test shall be "endurance run". The reliability test shall be performed on the two sets furnished specifically for the reliability test, when applicable. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.3.

4.3.4 Inspection report. The contractor shall furnish, within the time frame specified (see 6.2), an inspection report in accordance with MIL-STD-831 covering preproduction inspection. The inspecton report shall also include the following:

(a) All test data obtained from applicable tests.
(b) Description of malfunctions, damages, failures and adjustments (other than adjustments permitted by this specification) which occur during inspection.
(c) Causes and analysis of malfunctions, damages or failures and reasons for adjustments.
(d) Corrective action taken or required.

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4.4 Individual generator inspection. Inspection shall be performed prior to assembly into a set.

4.4.1 Examination. Each generator (with exciter-regulator system) shall be examined without disassembly for conformance to the drawings. Evidence that the generator (with exciter-regulator system) does not conform to the drawings shall be cause for rejection.

4.4.2 Tests. Before final assembly of the generator, each generator shall be subjected to the first two tests marked "X" under column 3 of table I. Failure of any test shall be cause for rejection.

4.5 Individual set inspection.

4.5.1 Examination. Each set shall be examined without disassembly for conformance to the drawings. Evidence that the set does not conform to the drawings shall be cause for rejection.

4.5.2 Tests. Each set shall be subjected to the tests marked "X" under column 3 of table I with new air, fuel, and oil filters installed. Failure of any test shall be cause for rejection.

4.6 Initial production inspection. When specified (see 3.3), one or more initial production sets will be selected at random by the Government from the sets being produced by production tooling and will be examined for conformance to the applicable drawings and subjected to the tests marked "X" in column 2 of table I to determine conformance to the requirements of this specification. These sets shall have successfully passed the examination and tests specified in 4.5. The inspection will be performed by the Government at a site selected by the Government, utilizing the appropriate technical manual for maintenance and servicing procedures. Acceptance of an initial production set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production tests, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.6.1 Inspection failure. Failure of an initial production set to meet any requirements specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the initial production sets and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all sets accepted prior to the completion of initial producton inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Quality conformance inspection.

4.7.1 Lots.

4.7.1.1 Generator with regulation system assemblies. Lots shall be formed from generators which have successfully passed the inspection specified in 4.4 and regulation system assemblies that have successfully passed the inspection specified in 4.1.1 and 4.1.2.

4.7.1.2 Sets. Lots shall be formed from generators which have successfully passed the inspection specified in 4.5. The first lot shall be not less than 25 sets if more than 25 are ordered.

4.7.2 Sampling for generator with regulation system assemblies and sets. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.7.3 Tests. Samples selected in accordance with 4.7.2 shall be subjected to the tests marked "X" in column 4 of table 1. AQL shall be 6.5 percent defective. No adjustment or substitution of components shall be made on the sets selected for sample inspection. These tests shall be performed utilizing the auxiliary fuel provision.

4.8 Test schedules.

4.8.1 Table I. The test schedule shall be as shown in table I. Nonconformance to the applicable requirement paragraph shown in column 7 of table I shall constitute failure of these tests. The Government reserves the right to reject the equipment for not meeting any requirement herein, even though not performing a test directly related to the specific requirement. The requirements in Section 3 and table I shall apply to all tests performed as a part of another test.

4.8.2 Table II. The optional test shall be as shown in table II. These tests shall be performed when required by the contracting officer to prove suitability and set compliance as a result of any changes in parts or material. Nonconformance to the applicable requirement paragraph shown in column 7 of table II shall constitute failure of these tests. The Government reserves the right to reject the equipment for not meeting any requirement herein, even though not performing a test directly related to the specific requirement. The requirements in Section 3 and table I shall apply to all tests performed as a part of another test.

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TABLE I Test schedule.

First Article (Preproduction)	Init Prod	Indiv	Qual Conf Sample	Test	Test Method (MIL-STD-705) Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
				GENERATOR ONLY		
X	-	X	-	Winding resistance	401.1 (Measure all windings before installation of rectifiers using Kelvin or Wheatstone bridge methods)	3.9.2
X	-	X	-	High potential	302.1 (Test each winding before installation of rectifiers.)	3.9.5
				GENERATOR WITH EXCITER-REGULATOR SYSTEM		
X	-	-	X	Open-circuit saturation curve	410.1 (One point at rated voltage for quality conformance.)	3.9.14
X	-	-	-	Synchronous Impedance Curve (Short-circuit saturation curve)	411.1 (Measure exciter field, current and voltage instead of generator field.)	
X	-	-	X	Generator power input	415.1 (120/208V, 3 phase only.)	3.9.15
X	-	-	X	Rated load current saturation curve	413.1 (One point at rated voltage for quality conformance.)	3.9.14
X	-	-	-	Direct-axis transient reactance	425.1	3.9.7
X	-	-	-	Negative sequence reactance and impedance	422.1 (120/208V, 3 phase only.)	3.9.8

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TABLE I Test schedule (cont'd).

1	2	3	4	5	6	7
X	-	-	X	Short-circuit (mechanical strength)	625.1 (Perform for all possible line-to-line, line-to-neutral, and 3 phase conditions)	3.9.6
X	-	-	X	Overspeed (generator only)	505.3	3.9.1
				GENERATOR SET		
X	-	-	X	Insulation resistance	301.1 and 4.9.5 (stator only)	3.9.3
X	-	-	X	Voltage modulation	602.1 (At all voltage connections at no load, rated kW at 0.8 pf, and rated kW at 1.0 pf.) (See note 1)	3.9.13.8
X	X	-	-	Railroad impact	740.5 (see 4.9.4)	3.4.3
X	X	-	-	Humidity	711.1	3.4.4, 3.9.3
X	-	-	-	Dc control	655.1 (Paragraph 655.1.3.2 (k) shall be accomplished by resistively loading the battery charging system, with the set batteries disconnected.)	3.11
X	-	X	-	Frequency and voltage regulation stability and transient response (short term)	608.1 (For individual sets, 2 hours operation at rated load will be considered as stabilizing the set with field and temperature readings not being taken.)	3.8.3, 3.9.13
-	-	X	-	Maximum power for Gasoline and Diesel Sets (Production Sets)	640.4 (125 percent rated kW. @ 120/208V, 3 phase, 1.0 pf.)	3.5
X	-	X	-	Overspeed (set)	505.1 (At 2500 rpm for 5 minutes.)	3.4.2
X	-	X	-	Phase balance (voltage)	508.1 (120/208V, 3 phase.)	3.9.11

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TABLE I Test schedule (cont'd).

1	2	3	4	5	6	7
X	X	-	-	Circuit interrupter (short circuit)	512.1 (All voltage connections.)	3.10
X	X	-	-	Circuit interrupter (overload current)	512.2 (All voltage connections for First Article (Preproduction Model.))	3.10
-	-	X	-	Indicating Instrument (electrical)	513.1	3.12, 3.9.13.7
X	-	-	-	Indicating instrument (electrical)	513.1 (All voltage connections.)	3.12, 3.9.13.7
X	-	-	X	Voltage waveform (Harmonic analysis)	601.4 (At no load and rated kW, 1.0 and 0.8 pf. at each line-to-line, and line-to-neutral conditions all voltage connections) (See note 1)	3.9.9
X	-	-	X	Voltage dip for low power factor loads	619.1 (120/208V, 3 phase.)	3.9.13.6
X	-	-	-	Voltage dip and rise for rated load	619.2 (120/208V, 3 phase and 120V, 1 phase.)	3.9.13.5
X	-	-	-	Voltage unbalance with unbalanced load (line-to-line)	620.2 (120/208V, 3 phase only.)	3.9.10
X	-	-	X	Low oil pressure protective device	515.1	3.8.4.2
X	-	-	X	Over-temperature protective device	515.2	3.8.4.1
X	-	-	-	Lifting and towing	740.4	3.4.5
X	-	X	-	Phase sequence (rotation)	507.1 (120/208V, 3 phase)	3.9.12
X	-	-	-	Maximum power (for Gasoline and Diesel Generator Sets)	640.1 (125 percent rated kW @120/208V, 3 phase, 1.0 pf.)	3.5

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TABLE I Test schedule (cont'd).

1	2	3	4	5	6	7
X	-	-	X	Frequency adjustment range	511.2	3.8.3.6
X	-	-	-	High temperature (125° F)	710.1 (All voltage connections) (Take temperature measurements on stator windings only.)	3.4.1
X	X	-	-	Drop (Freefall)	740.2 (Height, 6 inches flat position.)	3.4.3
X	X	-	-	Drop (Ends)	740.3 (Height, 12 inches.)	3.4.3
-	-	-	X	High temperature (125° F)	710.1 (Measure temperature rise at the condition which gave the highest rise in the stator windings as determined in the first article high temperature test.)	3.4.1
X	X	-	-	Endurance	690.1 (Shall be for 500 hours. Omit 690.1.3.2.3 (b), (c). Maintenance shall be performed in accordance with table III. Perform 608.1 at every 100 hours.)	3.4.1
-	-	-	X	Endurance	690.1 (Shall be for 200 hours.)	3.4.1

NOTE 1. For column 4 testing, perform at the 120/208V, 3 Ph connection at 0.8 pf only.

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TABLE II. Optional tests.

First Article (Preproduction)	Init Prod	Indiv	Qual Conf Sample	Test	Test Method (MIL-STD-705) Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	-	-	-	Reliability	695.1 (See 4.9.3, 4.9.3.1. Test Method 720.1 shall be performed before and after reliability test. Altitudes shall be 8000 feet (564.9 mm Hg), 95° F (35° C) and 5000 feet (632.4 mm Hg) 107° F (41.7° C). One set shall run 200 hours using JP4 fuel.)	3.4.1, 3.7
X	-	-	-	Road Test	4.9.6	3.4.3
X	-	-	-	Torsiographing	504.2	3.4.6
X	-	-	-	Sound level	661.2	3.14
X	-	-	-	Rain	711.3 (at 12 inches per hour)	3.4.1
X	-	-	-	Temperature drift	Contractor Prepared, Government Approved	3.8.3.4, 3.9.13.4
X	-	-	-	Safety		3.18
X	-	-	-	Human Engineering		3.18
X	X	-	-	Altitude operation	720.1 (At 8000 ft (564.9 mm Hg) and 95° F (35° C). Take temperature rise measurements on all stator windings)	3.4.1
X	X	-	-	Starting and operating (moderate cold battery start)	701.2 (At -25° F) (-31.7° C) (511.1 at all voltage connections.)	3.4.1
X	X	-	-	Inclined operation	660.1	3.4.1

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TABLE II. Optional tests. (cont'd)

1	2	3	4	5	6	7
X	X	-	-	Electro-magnetic interference	4.9.2	3.6
X	-	-	-	Starting and Operating (extreme cold battery start)	701.1 (At -50° F (-45.6° C))	3.4.1
X	-	-	-	Voltage Waveform (Oscillo-graphic)	601.1 (At no load and rated load 1.0 and 0.8 PF at each line-to-line and line-to-neutral conditions for all voltage connec-tions.)	3.9.9

TABLE III Servicing, adjustment and replacement schedule.

No.	Maintenance Interval (Hrs)	Servicing, Adjustment and Replacement Allowed
1	2	3
1.	As required	Clean air cleaner.
2.	8	Add lubricating oil, if necessary.
3.	8	Inspect engine for loose connections, leaks in oil and fuel systems, cracks and free action of all moving parts. Make minor adjustments.
4.	24	Fill batteries, inspect battery cable connections. Clean and grease as necessary.
5.	50	After the initial 50 hours of operation each engine shall have its valve tappet clearances adjusted.
6.	100	Change lubricating oil and filter.
7.	100	Change fuel oil filters.
8.	100	Clean crankcase breather.
9.	100	Adjust governor and throttle linkage, if necessary.
10.	500	Check valve tappet clearance and reset, if necessary.
11.	2500	Take compression pressure at cranking speed.
12.	2500	Clean or replace injection nozzles.
13.	2500	Replace governor ball joints.
14.	5000	Overhaul engine.

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4.9 Test procedures.

4.9.1 Test conditions. Test shall be conducted in accordance with MIL-STD-705 and as specified herein. Fuel conforming to VV-F-800, grades DF-1, DF-2, or DF-A as appropriate to the set environment, shall be used for all tests except the 200 hour JP-4 requirement during reliability. A new oil filter and new lubricants as specified below shall be used for each set during test.

SPECIFICATION	GRADE	TEMPERATURE
MIL-L-2104	30	Above 32° F
MIL-L-2104	10	-10° F to +32° F
MIL-L-46167	--	Below -10° F

Test instrumentation shall be in accordance with MIL-HDBK-705. Test instruments shall be of the laboratory type and shall have been calibrated within 30 days of the start of testing and at 6-month periods thereafter. Instruments used in calibration should have at least five times the accuracy of the instrument being calibrated. Calibrated reference instruments of lesser accuracy than standard and which are not used for any other purpose may be used for the required periodic check of test instruments. Instruments shall be calibrated at the frequencies at which they are going to be used. Direct-reading instruments shall have at least 0.5 percent instrument manufacturer's accuracy and shall be connected to indicate in the most accurate portion of their range. On dc instruments, the readings shall not be made on the lower 15 percent of the scale. On ac instruments, the readings shall not be made on the lower one-third of the scale. When the test methods call for the use of voltage and frequency recording-type meters, the Texas Instrument Company Model PDRHXFHXVA-A16-XT shall be used. Oscillograph frequency response shall be flat from dc to not less than 3,000 Hz. When recording meters are specified for any part of a test, turn on the recording meters prior to starting the warmup period of the set and record continuously for the duration of the test. Unless otherwise specified, the recording meters shall be operated at a minimum speed of 6 inches per hour during the portions of the test where steady-state loading conditions exist and shall be operated at a minimum speed of 6 inches per minute at least 30 seconds before, during, and after a load change. Individual tests requiring a series of data points shall be repeated if an interruption occurs during the test (e.g., data points being taken for plotting a curve, etc). Unless otherwise specified (see 6.2), all generators with exciter-regulator system tests shall be performed at 120/208 volt, 3 phase, and 120 volt, single phase connections. Unless otherwise specified in table I and II, all set tests shall be performed at the 120 volt single phase connection. During testing, the sets shall be positioned such that the exhaust of one unit does not directly impinge on another set.

4.9.2 Electromagnetic interference. The electromagnetic interference test shall be performed in accordance with MIL-STD-462. Tests shall be performed at all voltage connections at the following conditions:

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(a) No load.
(b) Rated kW at 1.0 pf.
(c) Rated load at 0.8 pf.

4.9.3 Reliability test. Test data shall be analyzed and evaluated in accordance with the failure definitions of 6.4.12. The accept-reject criteria shall be Test Plan III of MIL-STD-781.

4.9.3.1 Government-performed test. The reliability test shall be perfomed by the Government at its expense at a Government installation or at one designated by the Government.

4.9.4 Railroad impact. The railroad impact test shall be performed in accordance with method 740.5 with the following changes and information:

(a) Paragraphs 740.5.3(a), (c), (e), and 740.5.4(b): Delete the word "generator" and substitute "generator set".
(b) Paragraphs 740.5.3(b) and (f): Delete "Method 614.1, Voltage and Frequency Regulation Test." and substitute "Method 608.1, Frequency and Voltage Regulation, Stability and Transient Response Test on each set at rated load only."
(c) The generator sets shall be mounted with their longitudinal axis parallel to the length of the flat car. The sets shall not be packaged for the railroad impact tests. Each set shall be subjected to two impacts at 8, 9 and 10 miles per hour (± 0.5 mph) for a total of six impacts. Each of two sets shall be positioned 180 degrees with respect to each other. Anchorage of the set to the flatcar shall be as follows: Insert 6-inch steel channels through the forklift tine openings and securely bolt the channels to the flooring. Blocking timbers, front and rear, shall be used as necessary. Impact speed shall be measured within 60 inches of impact and the speed measurement interval shall not exceed 60 inches in length. Accuracy of the speed measurement shall be plus or minus 5 percent. Each set fuel tank shall be drained prior to impact of the sets. Drained batteries will be installed for impact test. All other liquids shall be at normal operating level.

4.9.5 Insulation resistance. Insulation resistance measurements may be made after assembly of the generator to the engine but before final assembly of the set.

4.9.6 Road test. The road test shall comprise four cycles of the following test schedule, with the set mounted on a suitable trailer:

(a) Paved highway - 250 miles at speeds up to 50 mph.
(b) Level cross-country - 250 miles at speeds up to 20 mph.
(c) Hilly cross-country - 125 miles at speeds up to 20 mph.
(d) Belgian block - 15 miles at speeds up to 20 mph.

4.10 Inspection comparison. The Government may select sets or components at any time during the contract production period and subject these sets or components to the examination and tests specified in 4.6 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on sets or components

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selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction models and initial production sets or components. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.10.1 Inspection failure. Failure of an inspection comparison or set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.10 shall be cause for rejection of the inspection comparison component or set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.11 Audio noise test. Except as specified herein, the instrumentation and procedures for the audio noise test shall conform to MIL-STD-1474. When any component is changed which may affect the noise limits (see 3.14) the number of feet to be specified in the caution notice as required in MIL-STD-1474 shall be determined by selecting the measurement position with the highest sound pressure levels and repositioning the microphone outward to a point where the measured sound pressure levels fall within the specified limits. Measure the distance from the set to the new microphone position and repeat the test for the eight positions at the new distance. All tests shall be conducted at no-load and rated load.

4.12 Inspection of packaging. Inspection of packaging shall be in accordance with the Quality Assurance Provisions of MIL-G-28554.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with MIL-G-28554. Preservation and packing shall be level A, B, or C as specified (see 6.2).

6. NOTES

6.1 Intended use. The generator sets specified herein are intended to supply power for general purpose use in military field applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification, and of the applicable specification sheet.
(b) Size of set required (see 1.2).

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(c) Time frame required for submission of the first article (preproduction model) (see 3.2).
(d) Whether two additional first article (preproduction model) sets for performing the reliability test are required (see 3.2).
(e) Whether initial production inspection is required and number of sets to be furnished (see 3.3).
(f) Whether sets are to be shipped connected for voltage other than as specified (see 3.13).
(g) Time frame for submission of the first article (preproduction) inspection report (see 4.3.4).
(h) When all system tests shall be performed other than as specified (see 4.9.1).
(i) Degree of preservation and degree of packing required (see 5.1).
(j) Whether sets are to contain a slave receptacle (see 3.15).
(k) Whether sets are to contain a ground rod assembly (see 3.16).

6.3 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the sets covered by this specification as contained in the the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91, when requested by the contractor.

6.4 Definitions.

6.4.1 Start. A set is considered to have started when it is operating at rated voltage and rated speed without the further use of starting aids.

6.4.2 Standby operation. A condition during which a nonoperating set is kept warm and ready to start through use of a winterization system, TA-30554-76-0750 or TA-30554-72-5980.

6.4.3 Rated load. Rated load is rated kW at rated pf, rated frequency, and rated voltage.

6.4.4 Temperature and humidity damage. Temperature and humidity damage is defined as any condition causing malfunction of any component or part, corrosion, breakage, deformation, reduction of insulation resistance below 50,000 ohms.

6.4.5 Rough handling damage. Rough handling damage is defined as any condition resulting in malfunctioning of the set, deformation, loosening, breakage, or change of fit of any component or part.

6.4.6 Normal railroad transportation. Normal railroad transportation shall be interpreted to mean impact speeds up to and including 10 miles per hour under the conditions specified in 4.9.4.

6.4.7 Normal truck or trailer transportation. Normal truck or trailer transportation is defined as the conditions encountered during the test specified in 4.9.6. This test can provide pulse inputs up to 20 g's along any axis, sustained unidirectional acceleration up to 1g along any horizontal axis, sustained sinusoidal vibration up to 4 g's or any combination or permutation of these conditions.

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6.4.8 Normal aircraft or helicopter transportation. Normal aircraft or helicopter transportation is defined as the conditions encountered during the 6-inch free fall drop and the 12-inch end drop under the test conditions specified in MIL-STD-705, Methods 740.2 and 740.3 respectively.

6.4.9 Frequency and voltage regulation. Frequency and voltage regulation is defined as the difference between the no load value and the rated load value divided by the rated load value. To express regulation as a percentage, multiply the value found by 100.

6.4.10 Stable set operating conditions (voltage and frequency). Stable set operating conditions are defined as the conditions specified for steady-state performance.

6.4.11 Meter error. Error is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.4.12 Failure. A failure during the reliability test is defined as the inability to perform within specified limits. The contracting officer will identify and classify all failures into the following categories.

6.4.12.1 Relevant failure. A relevant failure is any failure of an item which prevents the set from meeting the requirements specified herein. Relevant failures shall be used to compute set reliability.

6.4.12.2 Nonrelevant failure. Nonrelevant failures are those failures other than relevant failures. Examples of nonrelevant failures are as follows:

(a) Failure which does not prevent the set from meeting the power output requirements specified herein, e.g., a panel light burns out.
(b) Failure due to operator error where proper procedures are documented in technical manuals, instruction plates mounted on the set, or both; e.g., use of improper lubricant.
(c) Secondary failures caused by failures in the powered equipment or other occurrences in the set environment when integral set protection is not provided against such equipment failure or occurrence, e.g., explosion or fire.
(d) Failures due to characteristics of the load, e.g., waveform distortion caused by saturated inductors.
(e) Failures due to design deficiencies when subsequent testing demonstrates that the design deficiency has been corrected.
(f) Secondary failures caused by primary failures due to a design deficiency when subsequent testing demonstrates that the design deficiency has been corrected.
(g) Failures resulting from operating items beyond requirements, e.g., if ball joints are run to failure to determine mean life, failures after 2500 hours are not failures.

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6.4.12.3 Failure detection. A failure which would prevent the set from meeting any power output requirement specified herein is relevant notwithstanding the fact that it is detected during a shutdown for service and did not cause the set to perform out of limits during the previous testing (e.g., a shutter failure on a 40 degree day might not cause malfunction, but would preclude operation at the extreme temperatures and would be relevant, therefore, even if detected during shutdown for service).

6.4.13 Normal ambient conditions. Normal ambient conditions are considered to be at sea level with the ambient temperature between 60° F (15.6° C) and 85° F (29.4° C).

6.4.14 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.4.15 Examination. Examination is defined as an element of inspection that includes chemical, physical, and dimensional checks on the complete item including components and subassemblies and visual and tactile investigations.

6.4.16 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from flexural vibrations exceeds 9,000 psi.

6.4.17 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibration stress in the shaft from torsional vibration exceeds 5,000 psi.

6.4.18 Maintenance ratio. The maintenance ratio (MR) is the ratio of the number of manhours required for all scheduled and unscheduled maintenance per hour of set operation. Manhours at the operator/crew, organization, and direct support levels are included but not manhours to repair components or perform daily operational checks.

6.5 First article sets. Any changes or deviations of production sets from the approved first article sets (preproduction model) during production will be subject to the approval of the contracting officer. Approval of the first article (preproduction model) will not relieve the contractor of his obligation to furnish sets conforming to this specification.

6.6 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

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6.7 Recycled material. It is encouraged that recycled material be used when practical provided the material meets the requirements of the specification and all associated drawings.

6.8 Design. The design of the sets described by this specification permits storage at temperatures to minus 65° F (minus 53.6° C).

6.9 Configuration management. The generator sets controlled by this specification are under the Configuration Management authority of the DOD Project Manager for Mobile Electric Power. All Engineering Change Proposals (ECP), Requests for Waivers (RFW), and Request for Deviations (RFD) must be processed in accordance with DLAR 4120.7, AR 700-101, AFR 400-50, NAVMATINST 4120.100A, MC011310.8C, Joint Operating Procedures, Management and Standardization of Mobile Electric Power Generating Sources.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Navy - MC
DLA - GS

User activity:
Navy - CG

Preparing activity:
Army - ME

Project 6115-0227

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-G-52889C Generator Sets, Diesel Engine Driven 5 KW, 60 Hertz and 10 KW, 60 and 400 Hertz Tactical, Utility, Gen Spec for

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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